

Auxiliary equation

Basic skills

Q1. a) $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} - 20y = 0$

$a=1, b=5$
 $c=-20$

$1m^2 + 5m - 20 = 0$

b) $\frac{d^2y}{dx^2} + 9y = 0$
 $m^2 + 9 = 0$

c) $\frac{d^2y}{dx^2} - 9y = 0$
 $m^2 - 9 = 0$

Q2. $y = e^x, y' = e^x, y'' = e^x$

so $e^x - 3e^x + 2e^x = 0$ so yes

$y = e^{2x}, y' = 2e^{2x}, y'' = 4e^{2x}$

so $4e^{2x} - 3 \times 2e^{2x} + 2e^x = 0$ so yes

so solution is

$y = Ae^x + Be^{2x}$

for any constants A, B

any multiples
added
together

Q3. $y = e^{mx}$, $y' = me^{mx}$, $y'' = m^2 e^{mx}$

so $m^2 e^{mx} - 5me^{mx} - 6e^{mx} = 0$

As $e^{mx} \neq 0 \Rightarrow m^2 - 5m - 6 = 0$

Competency skills

Q4. For $y = e^{mx}$ to solve ODE
must have

$$am^2 e^{mx} + bme^{mx} + ce^{mx} = 0$$

and as $e^{mx} \neq 0 \Rightarrow am^2 + bm + c = 0$

in order for $y = e^{mx}$ to solve

Q5. As $am^2 + bm + c = 0$

is quadratic can have:

- 2 real (distinct)
- 1 real (repeated)
- 2 complex (distinct)

Q6.

a) 2 real m_1, m_2 then

$$y = Ae^{m_1 x} + Be^{m_2 x}$$

b) 1 real m then

$$y = Ae^{mx} + Bxe^{mx}$$

c) 2 complex $m = R \pm Ii$ then

$$y = e^{Rx} (A \cos(Ix) + B \sin(Ix))$$

Q7. $2 \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 2y = 0$

auxiliary: $2m^2 + 5m + 2 = 0$

$$\Rightarrow (2m+1)(m+2) = 0 \Rightarrow m = -\frac{1}{2}, -2$$

so 2 real roots so

$$y = Ae^{-\frac{1}{2}x} + Be^{-2x}$$

Q8. a) $m = R \pm Ii$ then as trying $y = e^{mx}$ have $y = Ae^{(R+Ii)x}$ and $y = Be^{(R-Ii)x}$ as solutions so have

$$y = Ae^{(R+Ii)x} + Be^{(R-Ii)x}$$

b) As Euler's identity: $e^{i\theta} = \cos\theta + i\sin\theta$

$$y = e^{Rx} \left(Ae^{(Ix)i} + Be^{-(Ix)i} \right)$$

↓ Euler ↓

$$= e^{Rx} \left(A(\cos Ix + i\sin Ix) + B(\cos(-Ix) + i\sin(-Ix)) \right)$$

As $\cos(-\theta) = \cos\theta$, $\sin(-\theta) = -\sin\theta$

$$= e^{Rx} \left((A+B)\cos Ix + (A-B)i\sin(Ix) \right)$$

so $C = A+B$, $D = (A-B)i$

Q9.

a) Initially trying $y = e^{mx}$ gives auxiliary so can only say this solves it at moment so $y = Ae^{mx}$ for any A

$$b) y = f(x)e^{mx} \quad y' = f'(x)e^{mx} + mf(x)e^{mx} \\ y'' = f''(x)e^{mx} + 2mf'(x)e^{mx} + m^2f(x)e^{mx}$$

$$\text{so } a(f''(x)e^{mx} + 2mf'(x)e^{mx} + m^2f(x)e^{mx}) \\ + b(f'(x)e^{mx} + mf(x)e^{mx}) + cf(x)e^{mx} = 0$$

$$\Rightarrow (af''(x) + 2amf'(x) + bf'(x))e^{mx} \\ + \underbrace{(am^2 + bm + c)f(x)e^{mx}}_{=0} = 0$$

$$\Rightarrow (af''(x) + (2am + b)f'(x))e^{mx} = 0$$

$\Rightarrow$ as $e^{mx} \neq 0 \Rightarrow af''(x) + (2am+b)f'(x) = 0$

$$\Rightarrow f''(x) + \left(2m + \frac{b}{a}\right)f'(x) = 0$$

Now $m = -\frac{b}{2a}$ as repeated roots

So $\Rightarrow f''(x) = 0$

$$\Rightarrow f'(x) = k \quad \text{only linear}$$

$$\Rightarrow f(x) = kx + C$$

So $y = (kx + C)e^{mx}$ solves

This is equivalent to

$$y = Ae^{mx} + Bxe^{mx}$$